Number Codes (Set B)

Year 4 · **Lesson Worksheet**

Worked Examples

Example 1: A number lock uses this code: 2 = H, 5 = O, 8 = T. What is the code for 258?

Answer: HOT — 2 = H, 5 = O, 8 = T, so 258 → H, O, T.

Example 2: A code is formed by doubling a number and then adding 1. What is the code for 7?

Answer: 15 — 7 × 2 = 14, then 14 + 1 = 15.

Example 3: In a code, every digit has 2 added to it, wrapping back round to 0 after 9 (so 9 becomes 1). What is the code for 793?

Answer: 915 — 7 → 9, 9 → 11 → 1, 3 → 5, giving 915.

Purple — Accessible (15 questions)

1. A suitcase lock uses this code: 3 = K, 6 = P, 8 = T, 9 = R. What is the code for the number 368?
2. A walkie-talkie uses this code: 1 = M, 4 = S, 5 = L, 7 = D. The code DLS stands for which number?
3. A safe uses this code: 1 = L, 3 = O, 5 = C, 7 = K. What is the code for the number 357? A) OKC B) OCK C) COK D) KOC E) CKO
4. A treasure map uses this code: 2 = G, 3 = O, 5 = N, 9 = V. What is the code for the number 293?
5. A robot's control panel uses this code: 0 = Z, 4 = Q, 6 = X, 8 = H. The code XQH stands for which number?
6. A bike lock uses this code: 2 = M, 4 = I, 6 = X, 8 = T. The code MIT stands for which number? A) 284 B) 428 C) 824 D) 248 E) 482
7. A pirate's chest uses this code: 1 = A, 2 = B, 7 = Y, 9 = Z. What is the code for the number 192?
8. A spy radio uses this code: 3 = F, 4 = J, 5 = K, 6 = W. The code JWK stands for which number?
9. Using the code 0 = G, 1 = T, 2 = O, 3 = P, 4 = N, 5 = W, 6 = F, 7 = S, 8 = L, 9 = M, select ALL the numbers below whose code would begin with the letter 'T'. A) 184 B) 253 C) 102 D) 719 E) 136
10. A garage door uses this code: 0 = B, 3 = E, 5 = D, 9 = G. What is the code for the number 903? A) GBE B) BGE C) EBG D) GEB E) BEG
11. A vending machine uses this code: 0 = C, 2 = E, 5 = I, 7 = U. What is the code for the number 207?
12. A garden gate keypad uses this code: 1 = T, 3 = R, 6 = E, 8 = N. The code NET stands for which number?
13. A phone passcode uses this code: 1 = P, 2 = A, 4 = S, 6 = T. The code SAP stands for which number? A) 124 B) 214 C) 421 D) 412 E) 241
14. A castle drawbridge uses this code: 2 = D, 4 = R, 5 = A, 9 = W. What is the code for the number 954?
15. A school locker uses this code: 1 = S, 2 = U, 3 = N. The code SUN stands for which number?

Green — Standard (15 questions)

1. A code is formed by doubling a number and then adding 5. What is the code for 9?

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2. A code is formed by multiplying a number by 3 and then subtracting 2. The code for a certain number is 22. What was the original number?

3. A code is formed by doubling a number and then subtracting 3. What is the code for 8? A) 11 B) 12 C) 13 D) 14 E) 16

4. A code is formed by adding 4 to a number and then doubling the result. What is the code for 6?

5. A code is formed by halving a number and then adding 7. The code for a number is 12. What was the original number?

6. A code is formed by multiplying a number by 4 and then adding 2. The code for a number is 30. What was the original number? A) 7 B) 8 C) 6 D) 9 E) 5

7. In a code, 3 → 11 and 5 → 19. The same rule is used for every number. What is the code for 7?

8. In a code, 4 → 14 and 9 → 24. The same rule applies to every number. If the code for a certain number is 20, what is that number?

9. A code is formed by multiplying a number by 3 and then adding 1. Select ALL the numbers below whose code would be greater than 20. A) 5 B) 6 C) 7 D) 8 E) 9

10. In a code, 5 → 17 and 8 → 26. Using the same rule, what is the code for 12? A) 34 B) 35 C) 36 D) 37 E) 38

11. A code is formed by subtracting 3 from a number and then multiplying the result by 3. What is the code for 10?

12. A code is formed by multiplying a number by 5 and then adding 1. The code for a number is 31. What was the original number?

13. A code is formed by subtracting 2 from a number and then multiplying by 4. What is the code for 9? A) 24 B) 28 C) 32 D) 20 E) 26

14. In a code, 2 → 8 and 6 → 20. What is the code for 10, using the same rule?

15. In a code, 3 → 8 and 7 → 16. If the code for a number is 12, what is that number, using the same rule?

Red — Challenge (15 questions)

1. In a code, every digit has 3 added to it, wrapping back to 0 after 9. What is the code for 142?

2. A code adds 3 to every digit, wrapping after 9. If a number's code is 609, what was the number before coding?

3. Add 5 to every digit (wrap after 9) to make a code. Which of these is the code for 274? A) 729 B) 719 C) 739 D) 629 E) 827

4. In a code, the digits of a number are simply reversed. What is the code for 583?

5. A number's digits are flipped end to end to make its code. If the code is 416, what was the number?

6. A code reverses the digits of a number. What is the code for 927? A) 729 B) 792 C) 279 D) 297 E) 972

7. In a code, every digit has 4 added to it, wrapping after 9. What is the code for 518?

8. Using the same rule (add 4, wrap after 9), the code for a number is 163. What was the original number?

9. Add 6 to every digit (wrap after 9) to make a code. Select ALL the numbers below whose code would include a 0 digit. A) 138 B) 249 C) 352 D) 461 E) 573

10. In a code, every digit has 7 added to it, wrapping after 9. What is the code for 206? A) 973 B) 972 C) 983 D) 963 E) 974

11. Using the same rule (add 7 to each digit, wrap after 9), the code for a number is 381. What was the original number?

12. In a code, every digit has 2 added to it, wrapping after 9. What is the code for 795?

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13. Using the same rule (add 2, wrap after 9), the code for a number is 134. What was the original number? A) 912 B) 921 C) 813 D) 802 E) 931

14. A code reverses the digits of a number. What is the code for 356?

15. Using the reversing rule, the code for a number is 481. What was the original number?

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